

Buffet Working Group

Test Case 1



Version 3
September 20, 2024

aiaabuffet@gmail.com



- **Geometry is available here: (it is very strongly desired to use the provided IGES file in the ONERA OAT15A zip file and not the raw coordinates)**

<https://aiaa-dpw.larc.nasa.gov/geometry.html>

- **Committee-supplied RANS grids are available here**

<https://aiaa-dpw.larc.nasa.gov/grids.html>

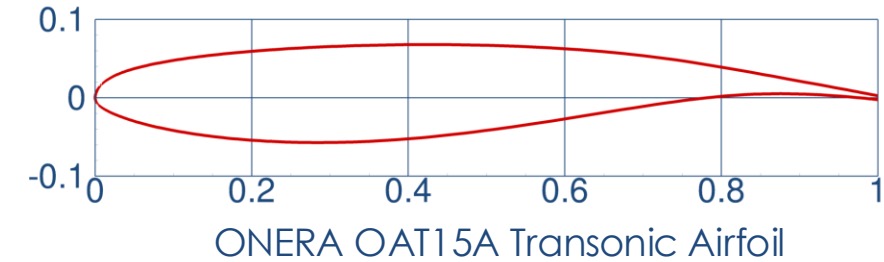
- **Experimental data are available here**

<https://aiaa-dpw.larc.nasa.gov/experiment.html>

Test Case 1a: Workshop-Wide Validation

- **Validation of steady CFD analysis, required**
- **Users are encouraged to employ best practices**
- **Settings**
 - Steady CFD (e.g., RANS)
 - Prefer some version of SA, multiple turbulence models can be submitted
 - Use periodic boundary conditions for sidewall boundary conditions
- **Grids**
 - Six-member grid family; four are required, six are desirable
 - Encourage use of committee-supplied grids; user-generated grids are acceptable
 - Three committee-supplied once-cell-wide grid topologies are provided
- **Conditions**
 - Mach 0.73, $Re_c=3m$ (based on chord length), $T_{static}=271\text{ K (487.8 R)}$
 - Alpha: 1.36, 1.50, 2.50, 3.00, 3.10, Buffet WG supplement: 3.25, 3.40, 3.50, 3.60, 3.90
 - Experimental conditions (for reference): $P_{total}=102.4\text{ kPa}$; $P_{static}=71.8\text{ kPa}$

Jaquin, et al. "Experimental Study of Shock Oscillation over a Transonic Supercritical Profiles." AIAA Journal, Vol. 47, No. 9, 2009. Pages 1985-1994.



Test Case 1a: Data Submission

- **Please follow these instructions**

<https://aiaa-dpw.larc.nasa.gov/postprocessing.html>

- **Required data**

- Forces and Moments

- DPW8-AePW4_ForceMoment_v5.dat

- Surface cuts

- DPW8-AePW4_SectionalCuts_v5.dat

- Use sectionalCutter-v2.mcr

- Convergence data

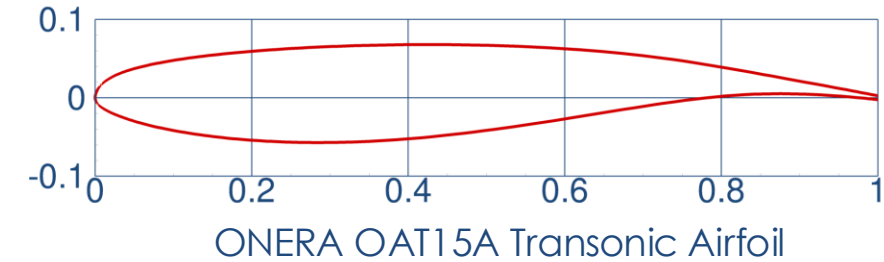
- DPW8-AePW4_Convergence_v5.dat

- Contour plots

- Use airfoilImages-v2.mcr

- **Validation of unsteady CFD analysis, required**
- **Users are encouraged to employ best practices**
- **Settings**
 - Unsteady CFD (e.g., URANS, DES, LES, etc.)
 - Prefer some version of SA, multiple turbulence models can be submitted
 - Use periodic boundary conditions for sidewall boundary conditions
- **Grids**
 - Same geometry options as Test Case 1a
 - Specialized grids for unsteady schemes will likely be generated by participants
- **Conditions**
 - Mach 0.73, $Re_c=3m$ (based on chord length), $T_{static}=271\text{ K (487.8 R)}$
 - Alpha: 1.36, 1.50, 2.50, 3.00, 3.10, 3.25, 3.40, 3.50, 3.60, and 3.90
 - Experimental (for reference) $P_{total}=102.4\text{ kPa}$; $P_{static}=71.8\text{ kPa}$

Jaquin, et al. "Experimental Study of Shock Oscillation over a Transonic Supercritical Profiles." AIAA Journal, Vol. 47, No. 9, 2009. Pages 1985-1994.



Test Case 1b: Data Submission (In Work)

- **Please follow these instructions**

<https://aiaa-dpw.larc.nasa.gov/postprocessing.html>

- **Required data**

- Forces and Moments

DPW8-AePW4_UnsteadyForceMoment_v5.dat

- Surface cuts

DPW8-AePW4_UnsteadySectionalCuts_v5.dat

- Spectral content

DPW8-AePW4_UnsteadySpectra_v5.dat



aiaabuffet@gmail.com